

•Fiber Optic Network Design

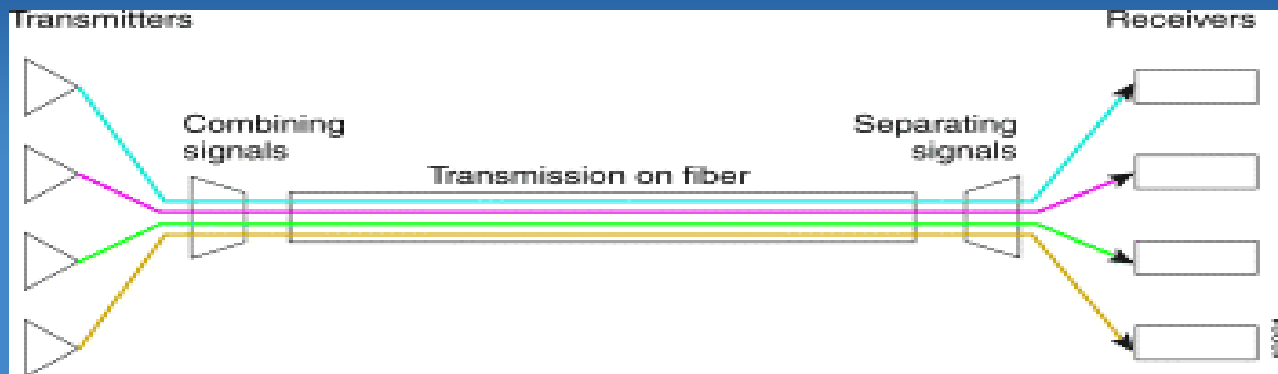
College of Opt. Sci. & Engr., ZJU
2026

•Outline

- WDM/DWDM history
- System Concerns
 - Channel spacing v.s. crosstalk
 - Other issues
- Exercise

What is DWDM?

- Definition
 - Dense wavelength division multiplexing (DWDM) is a fiber-optic transmission technique that employs light wavelengths to transmit data parallel-by-bit or serial-by-character

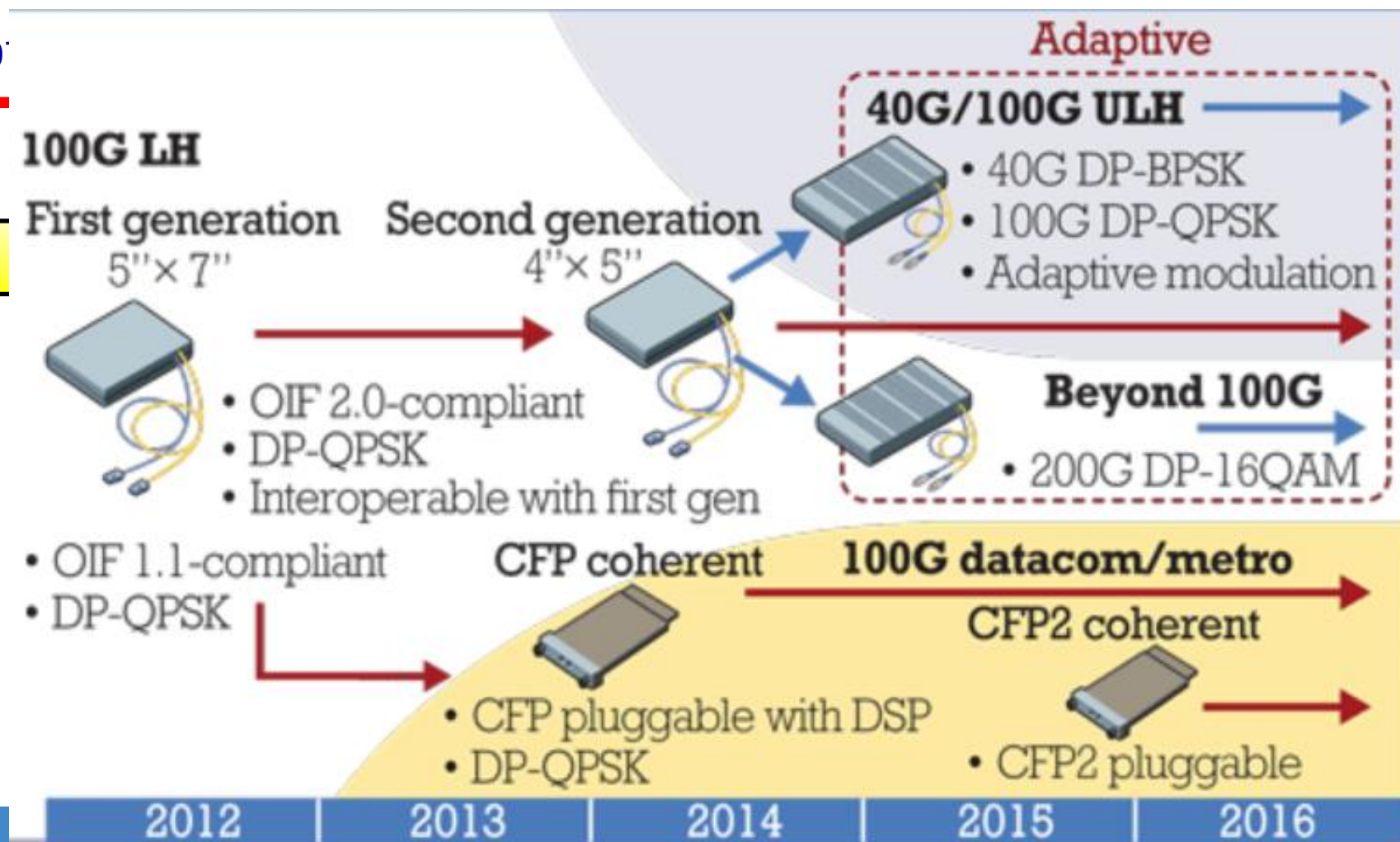


How does DWDM fair better?

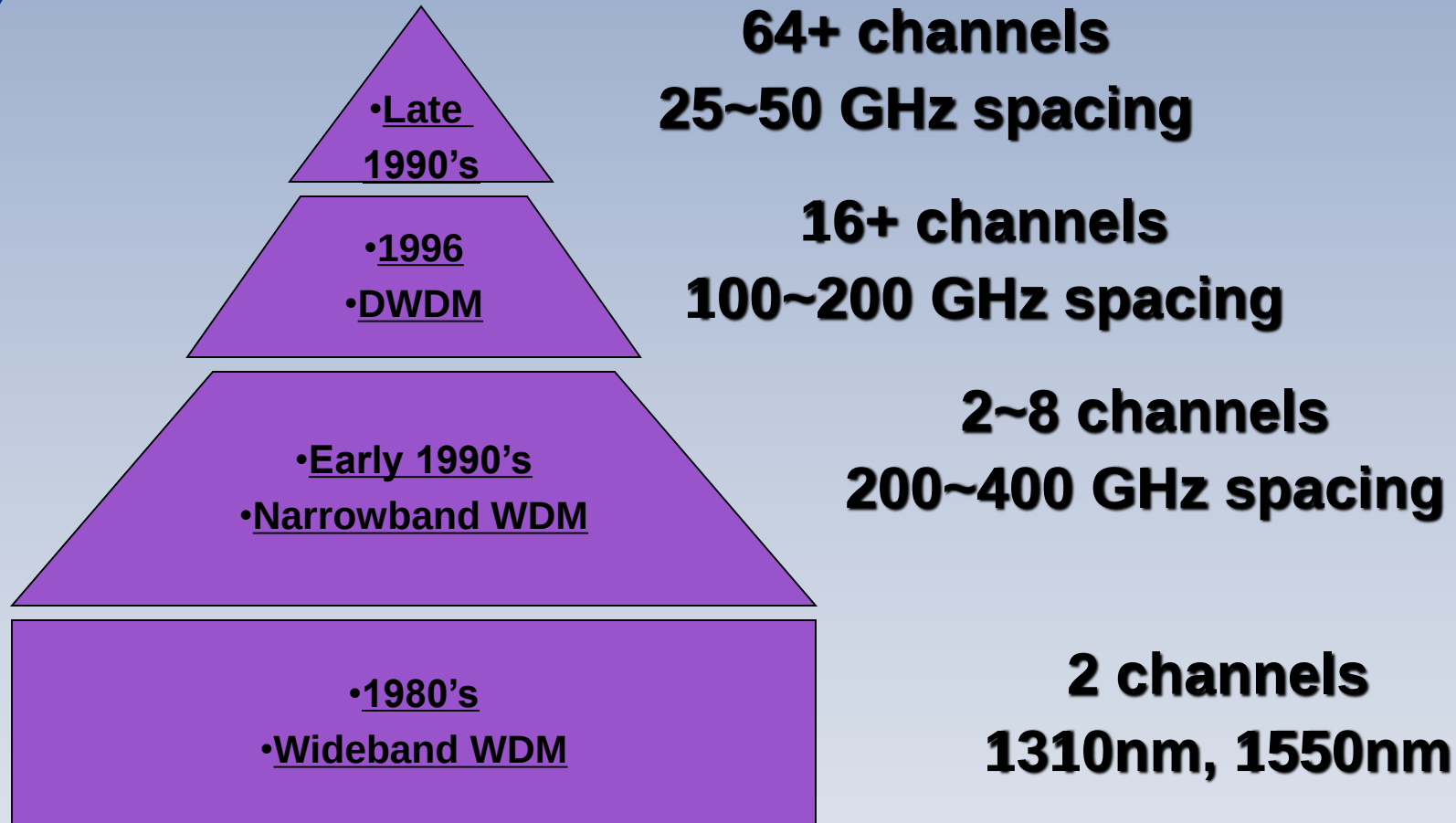
- No O-E-O required
- Protocol & Bit Rate independence
- Increased overall capacity at much lower cost
 - Current fiber plant investment can be optimized by a factor of at least 32
- Transparency
 - Physical layer architecture → supports both TDM and data formats such as ATM, Gigabit Ethernet, etc.
- Scalability
 - Utilize abundance of dark fibers in metropolitan areas and enterprise networks

Timeline

19



Evolution of DWDM



Problems and Solutions

Problem:

Demand for massive increases in capacity



Immediate Solution:

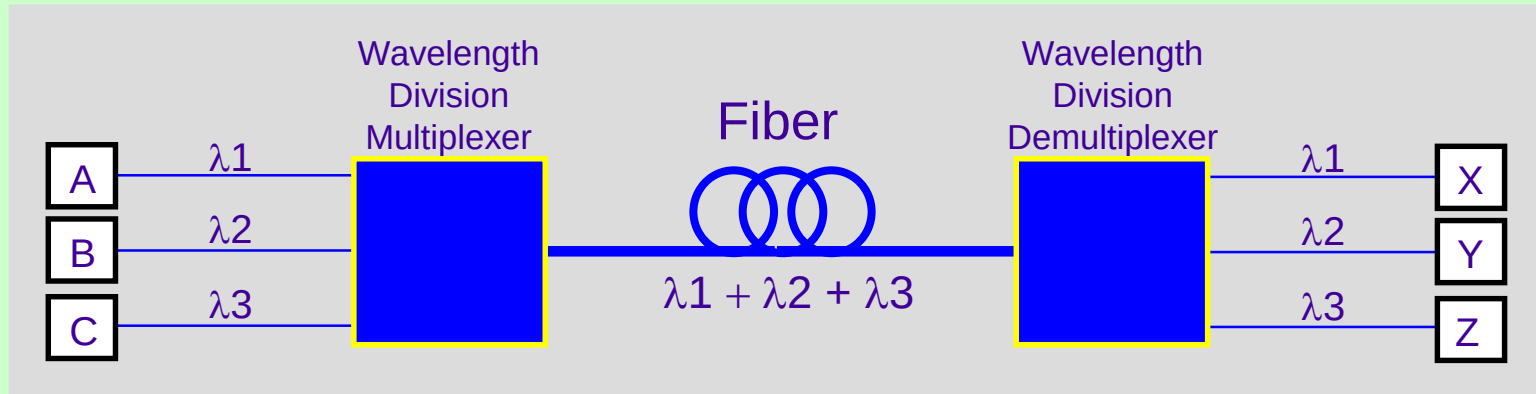
Dense Wavelength Division Multiplexing



Longer term Solution:

Optical Fibre Networks

WDM Overview



- Multiple channels of information carried over the same fibre, each using an individual wavelength
- Attractive multiplexing technique
 - High aggregate bit rate without high speed electronics or modulation
 - Low dispersion penalty for aggregate bit rate
 - Very useful for upgrades to installed fibres
 - Realizable using commercial components, unlike OTDM
- Loss, crosstalk and non-linear effects are potential problems

Types of WDM

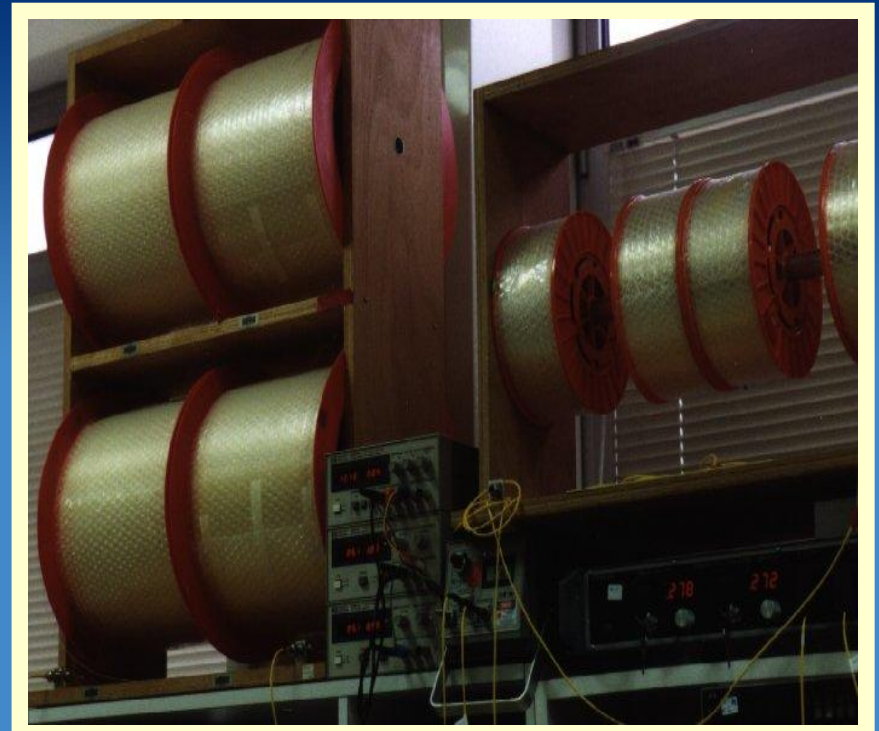
	Coarse WDM (includes WWDM)	WDM	DWDM (includes ultra dense WDM)
Channel Spacings	Large, from 1.6 nm (200 GHz) to 25 nm	1310 nm lasers used in conjunction with 1550 nm lasers	Small, 200 GHz and less
Number of bands used	O,E,S,C and L	O and C	C and L
Cost per channel	Low	Low	High
Number of channels delivered	17-18 at most	2	Hundreds of channels possible
Best Application	Short-haul, Metro	PON	Long-haul

Early DWDM: CNET WDM Experimental Setup

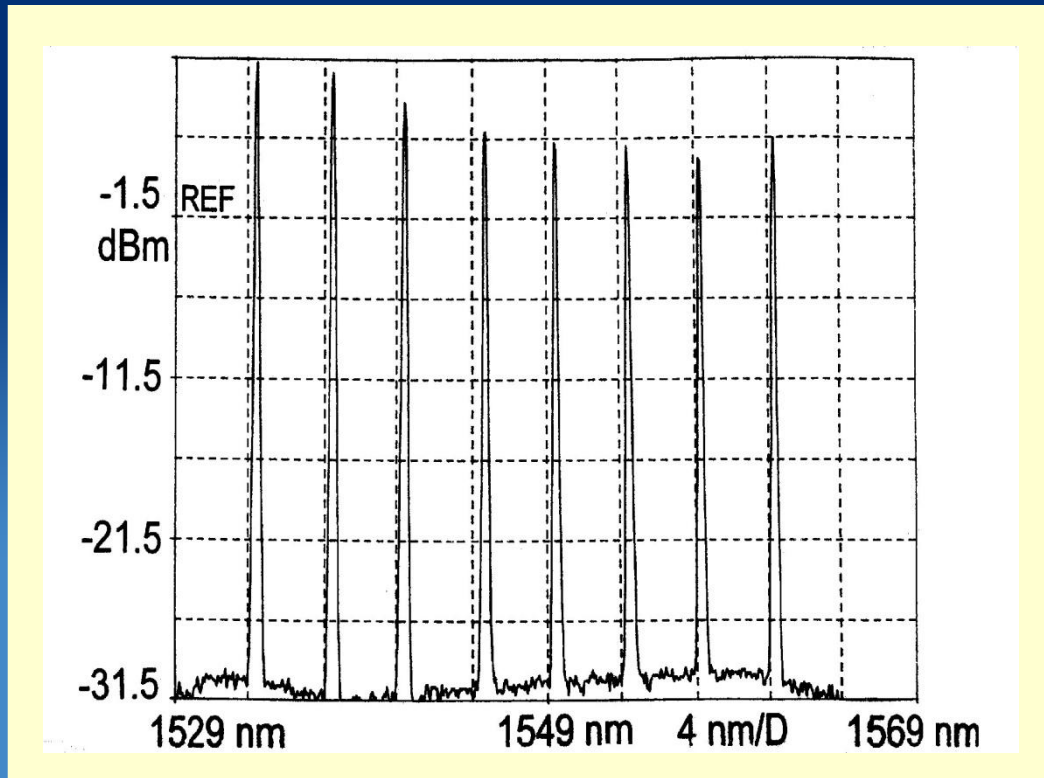


Optical Transmitters

Buffered Fibre on Reels



Sample DWDM Signal



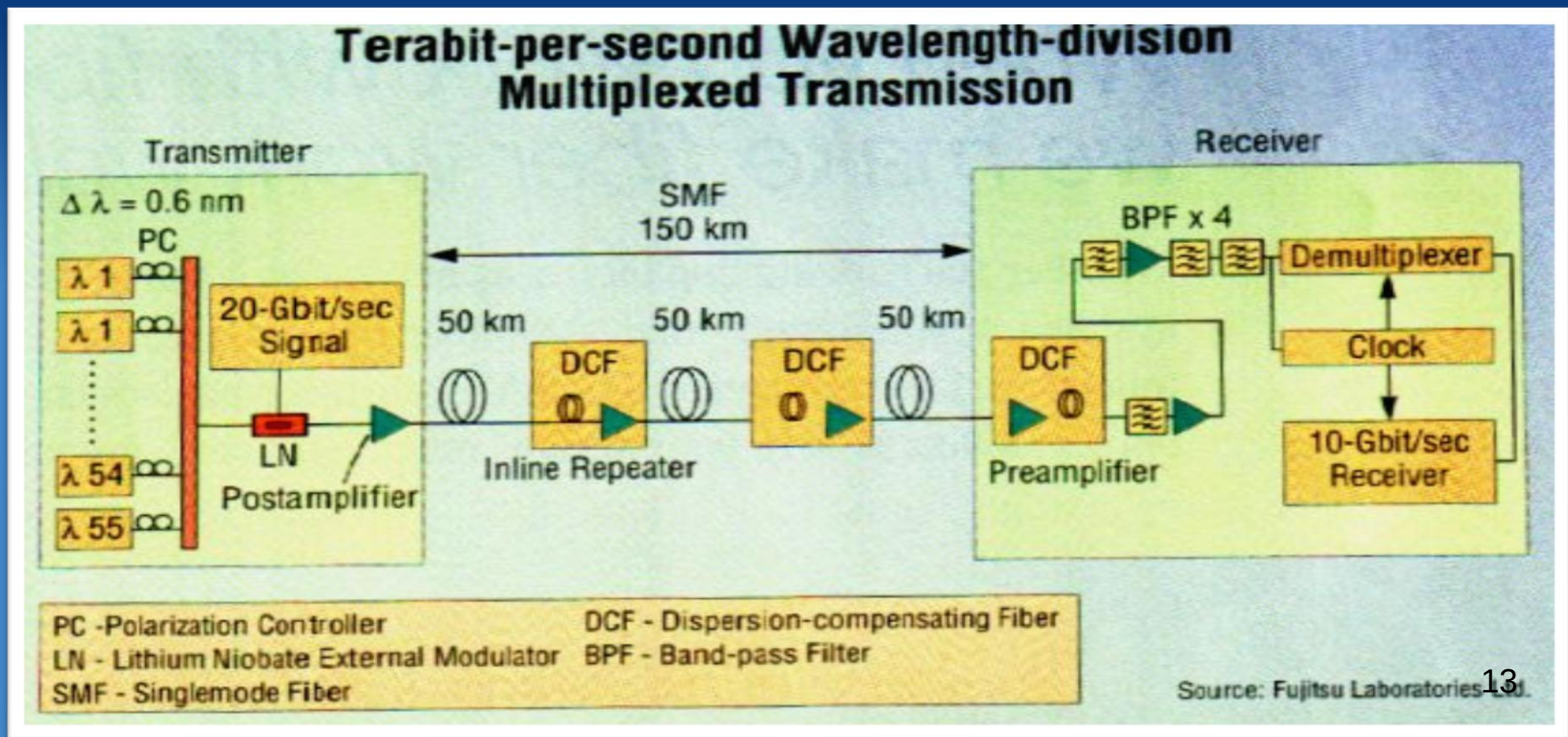
Multiplexer Optical Output Spectrum for an 8 DWDM channel system, showing individual channels

DWDM: Key Points

- *Dense WDM* is WDM utilising closely spaced channels
- Channel spacing reduced to 1.6 nm and less
- Cost effective way of increasing capacity without replacing fibre
- Commercial systems available with capacities of 32 channels and upwards; > 80 Gb/s per fibre
- Allows new optical network topologies, for example high speed metropolitan rings
- Optical amplifiers are also a key component

Terabit Transmission using DWDM

- 1.1 Tbits/sec total bit rate (more than 13 million telephone channels)
- 55 wavelengths at 20 Gbits/sec each
- 1550 nm operation over 150 km with dispersion compensation
- Bandwidth from 1531.7 nm to 1564.07 nm (0.6 nm spacing)



DWDM Advantages

- Greater fiber capacity
- Easier network expansion
 - No new fiber needed
 - Just add a new wavelength
 - Incremental cost for a new channel is low
 - No need to replace many components such as optical amplifiers
- DWDM systems capable of longer span lengths
 - TDM approach using STM-64 is more costly and more susceptible to chromatic and polarization mode dispersion

DWDM Disadvantages

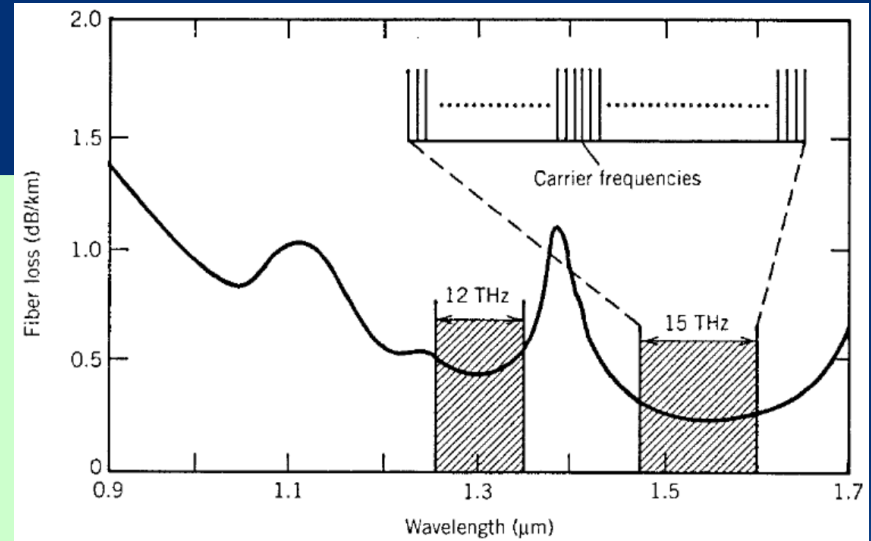
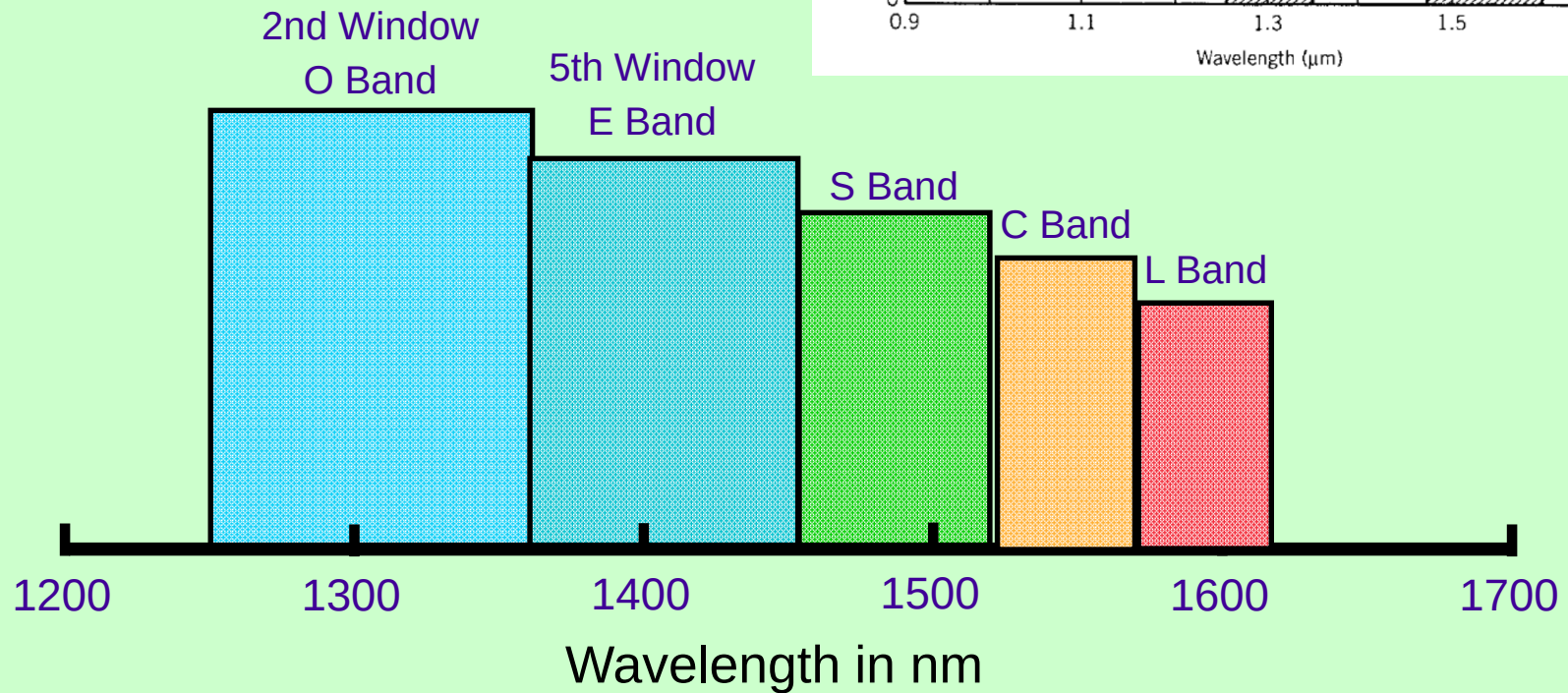
- Not cost-effective for low channel numbers
 - Fixed cost of mux/demux, transponder, other system components
- Introduces another element, the frequency domain, to network design and management
- SONET/SDH network management systems not well equipped to handle DWDM topologies
- DWDM performance monitoring and protection methodologies developing

DWDM Standards



- ITU Recommendation is G.692 "Optical interfaces for multichannel systems with optical amplifiers"
- G.692 includes a number of DWDM channel plans
- Channel separation set at:
 - 50, 100 and 200 GHz
 - equivalent to approximate wavelength spacings of 0.4, 0.8 and 1.6 nm
- Channels lie in the range 1530.3 nm to 1567.1 nm (so-called C-Band)
- Newer "L-Band" exists from about 1570 nm to 1620 nm
- Supervisory channel also specified at 1510 nm to handle alarms and monitoring

Optical Spectral Bands



ITU DWDM Channel Plan

0.4 nm Spacing (50 GHz)

All Wavelengths in nm

So called
ITU *C-Band*

81 channels defined

Another band called
the *L-band* exists
above 1565 nm

1528.77	1534.64	1540.56	1546.52	1552.52	1558.58
1529.16	1535.04	1540.95	1546.92	1552.93	1558.98
1529.55	1535.43	1541.35	1547.32	1553.33	1559.39
1529.94	1535.82	1541.75	1547.72	1553.73	1559.79
1530.33	1536.22	1542.14	1548.11	1554.13	1560.20
1530.72	1536.61	1542.54	1548.51	1554.54	1560.61
1531.12	1537.00	1542.94	1548.91	1554.94	
1531.51	1537.40	1543.33	1549.32	1555.34	
1531.90	1537.79	1543.73	1549.72	1555.75	
1532.29	1538.19	1544.13	1550.12	1556.15	
1532.68	1538.58	1544.53	1550.52	1556.55	
1533.07	1538.98	1544.92	1550.92	1556.96	
1533.47	1539.37	1545.32	1551.32	1557.36	
1533.86	1539.77	1545.72	1551.72	1557.77	
1534.25	1540.16	1546.12	1552.12	1558.17	

Speed of Light assumed to be 2.99792458×10^8 m/s

ITU DWDM Channel Plan

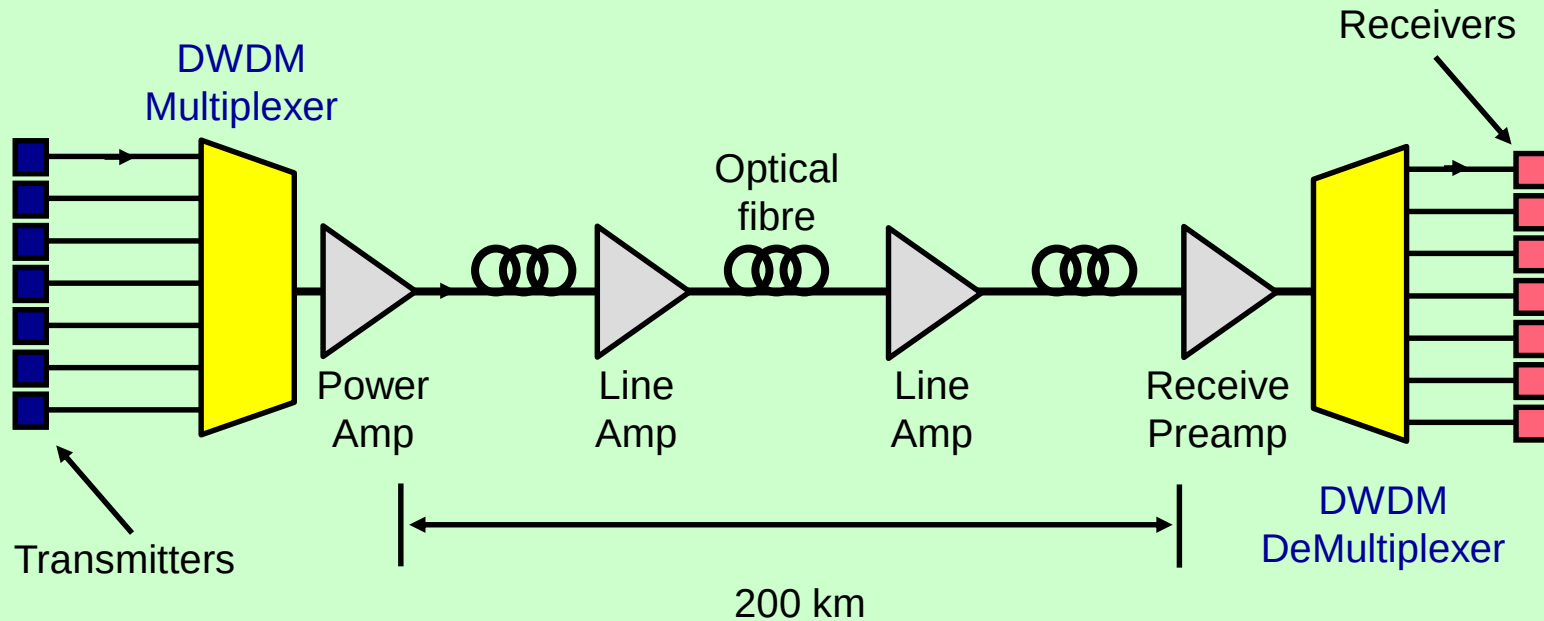
0.8 nm Spacing (100 GHz)

All Wavelengths in nm

1528.77	1534.64	1540.56	1546.52	1552.52	1558.98
1529.55	1535.43	1541.35	1547.32	1553.33	1559.79
1530.33	1536.22	1542.14	1548.11	1554.13	1560.61
1531.12	1537.00	1542.94	1548.91	1554.94	
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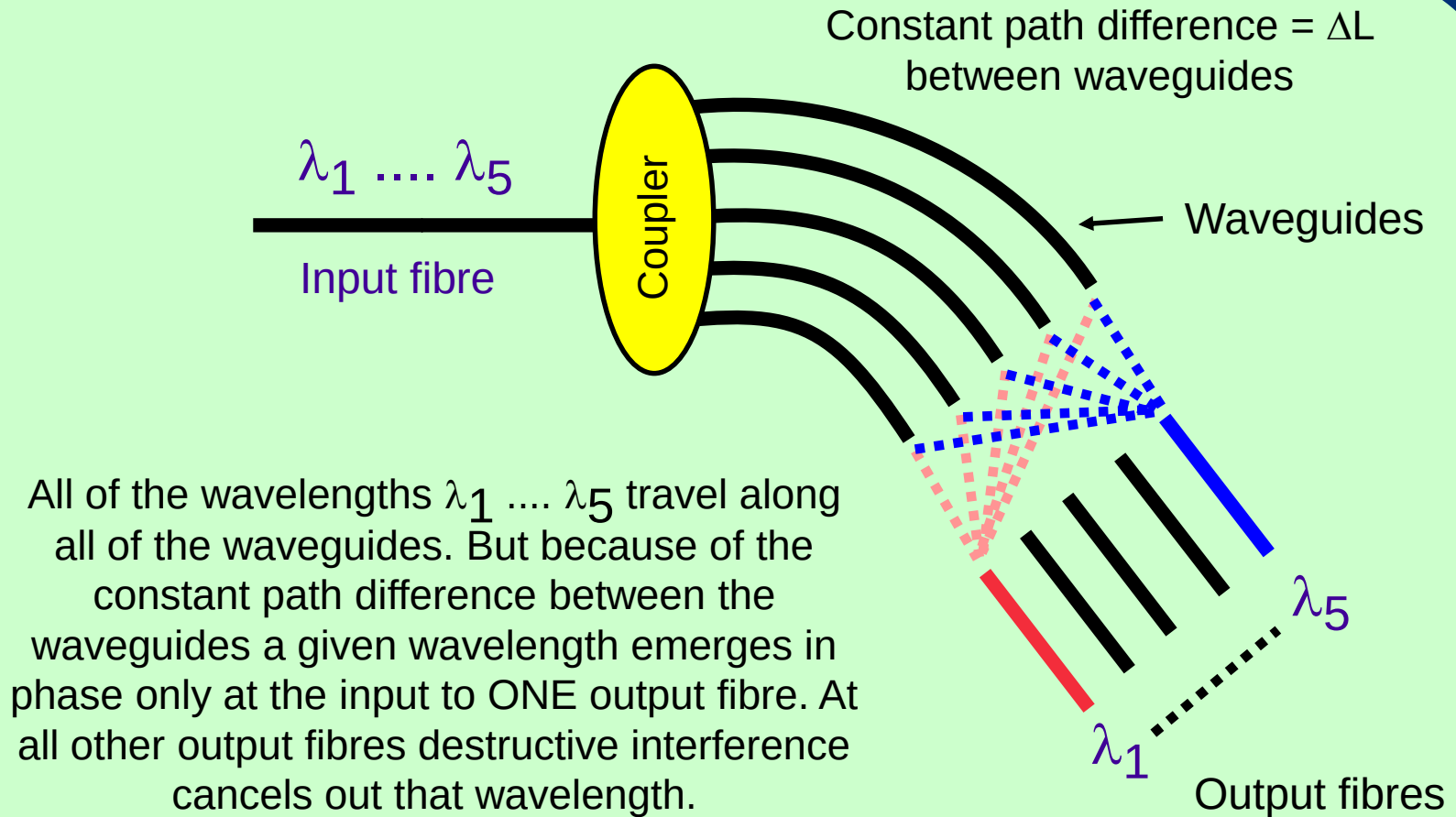
Speed of Light assumed to be 2.99792458×10^8 m/s

DWDM System



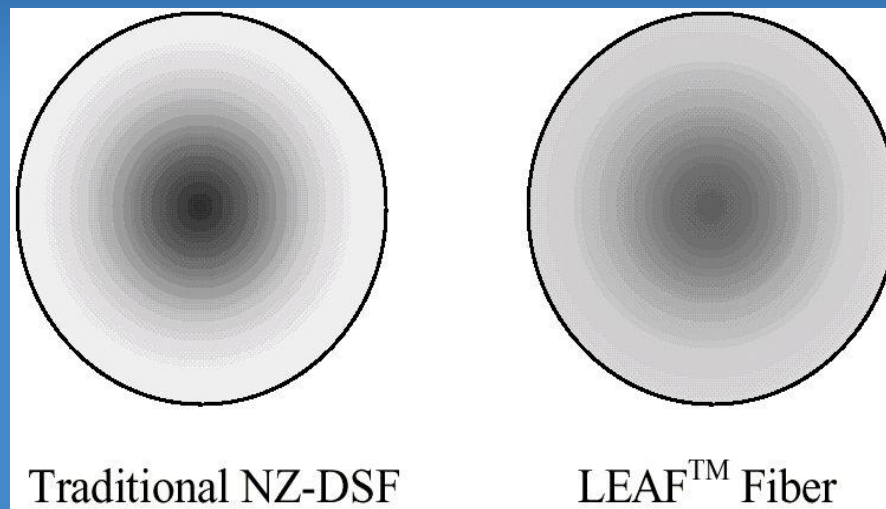
- Each wavelength behaves as if it has its own "virtual fibre"
- Optical amplifiers needed to overcome losses in mux/demux and long fibre spans

Array Waveguide Grating Operation: Demultiplexing



Reducing FWM using a Large Effective Area Fibre NZ-DSF

- One way to improve on NZ-DSF is to increase the effective area of the fibre
- In a single mode fibre the optical power density peaks at the centre of the fibre core
- FWM and other effect most likely to take place at locations of high power density
- Large effective Area Fibres spread the power density more evenly across the fibre core
- Result is a reduction in peak power and thus FWM



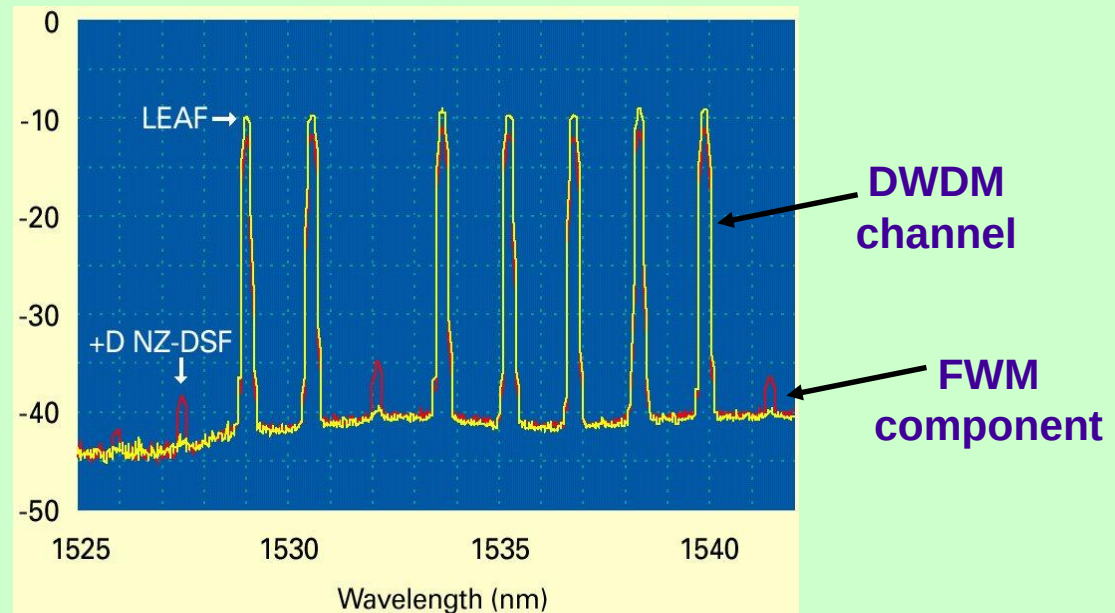
LEAF Fiber

- LEAF has an effective area 32% larger than conventional NZ-DSF
- Claimed result is lower FWM
- Impact on system design is that it allows higher fibre input powers so span increases

Section of DWDM spectrum

NZ-DSF shows higher FWM components

LEAF has lower FWM and higher per channel power



What is the 2011 record?

112-Tb/s Space-division multiplexed DWDM transmission with 14-b/s/Hz aggregate spectral efficiency over a 76.8-km seven-core fiber

B. Zhu,^{1,*} T.F. Taunay,¹ M. Fishteyn,¹ X. Liu,² S. Chandrasekhar,² M. F. Yan,¹ J. M. Fini,¹ E. M. Monberg,¹ F. V. Dimarcello¹

¹OFS Labs, 19 Schoolhouse Rd, Somerset NJ, 08873, USA

²Bell Labs, Alcatel-Lucent, 791 Holmdel-Keyport Road, Holmdel, NJ 07733, USA

*bzhu@ofsoptics.com

Abstract: We describe a new multicore fiber (MCF) having seven single-mode cores arranged in a hexagonal array, exhibiting low crosstalk among the cores and low loss across the C and L bands. We experimentally demonstrate a record transmission capacity of 112 Tb/s over a 76.8-km MCF using space-division multiplexing and dense wavelength-division multiplexing (DWDM). Each core carries 160 107-Gb/s polarization-division multiplexed quadrature phase-shift keying (PDM-QPSK) channels on a 50-GHz grid in the C and L bands, resulting in an aggregate spectral efficiency of 14 b/s/Hz. We further investigate the impact of the inter-core crosstalk on a 107-Gb/s PDM-QPSK signal after transmitting through the center core of the MCF when all the 6 outer cores carry same-wavelength

Today's Lab

- Setup a DWDM system
- Parameters
 - Fiber length
 - Channel spacing
 - BER/Q-factor
 - DCF

Design Target

- 40Gb/s 16 channel link between two cities A and B, spanning 300km
- $Q > 6$
- As few parts as possible
- Cost as low as possible

Design Procedure

- **Similar to writing a computer program:**
 - **Start from subroutine: make single channel 40G link work first**
 - **Make it DWDM**
 - **Know dependence of parameters:**
 - distance,
 - transmitter power,
 - in-line power,
 - channel spacing....

What type of fiber link to use to bridge two cities?

A



Optical Fiber 2
Length = 300 km

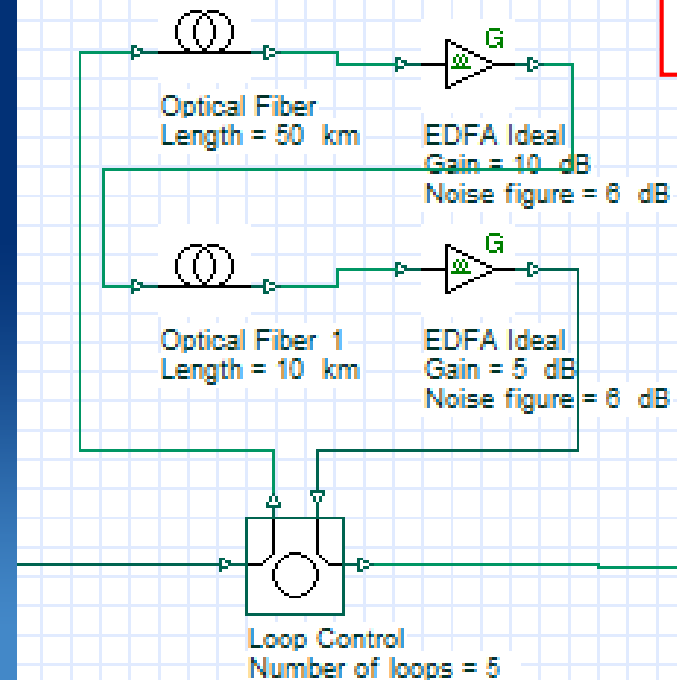
B



Optical Fiber 3
Length = 150 km

Gain=???
NF=???

C

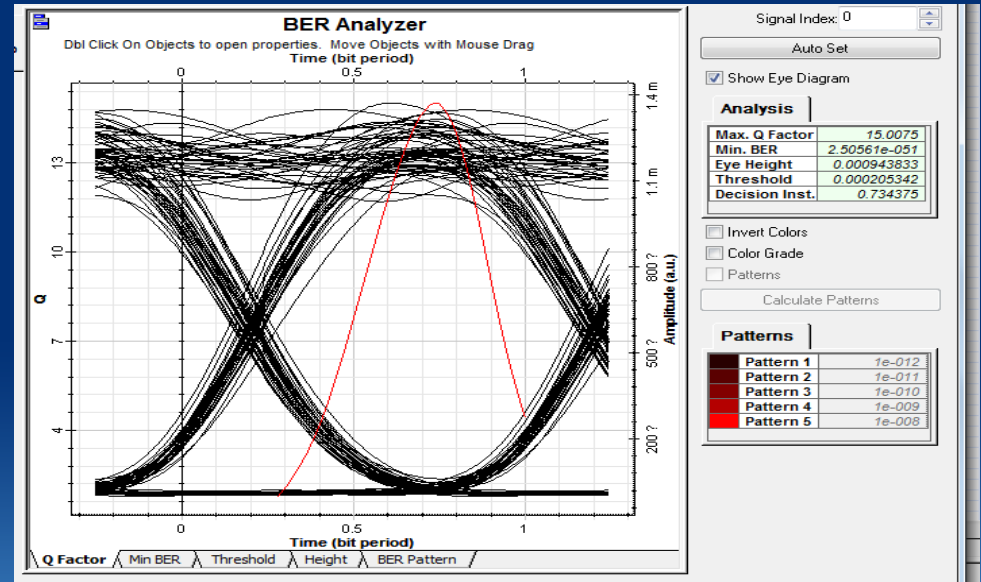


What is the general selection rule of fiber link?

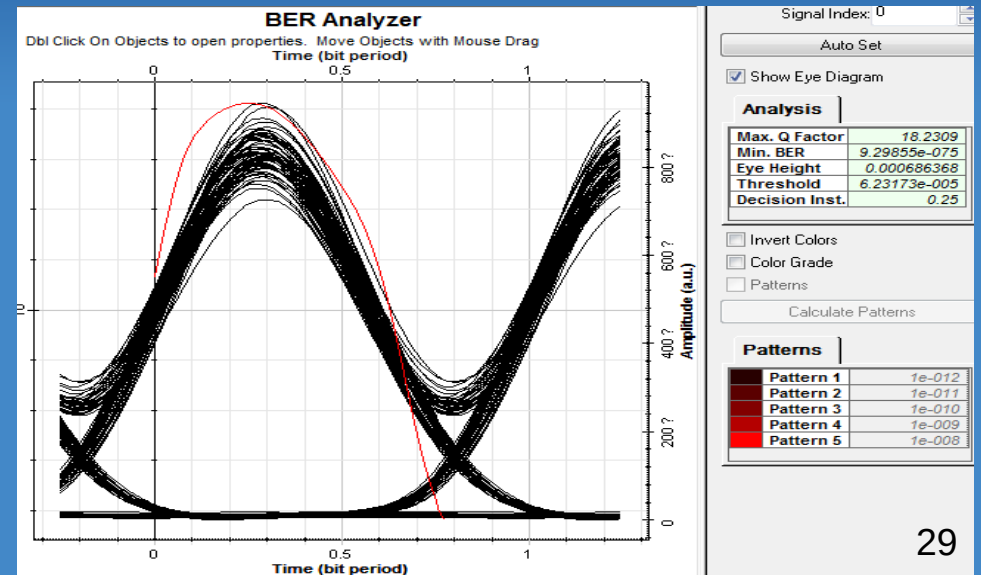
Post-compensation v.s pre-compensation

NRZ v.s. RZ, Which one to use?

NRZ: $Q=15$

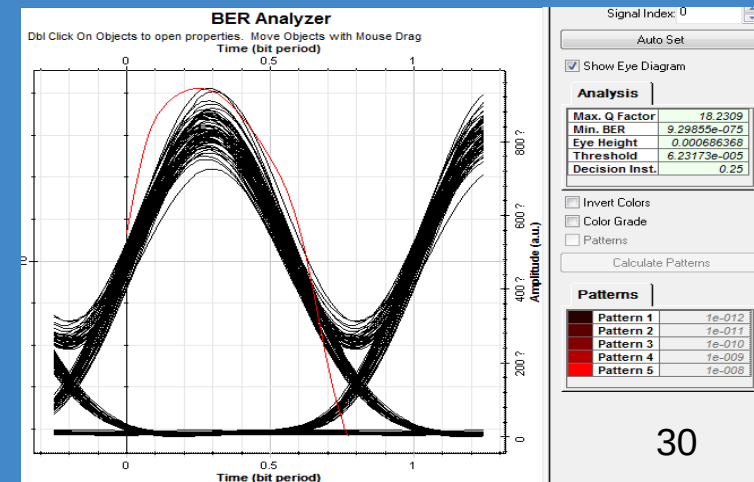
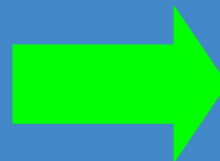
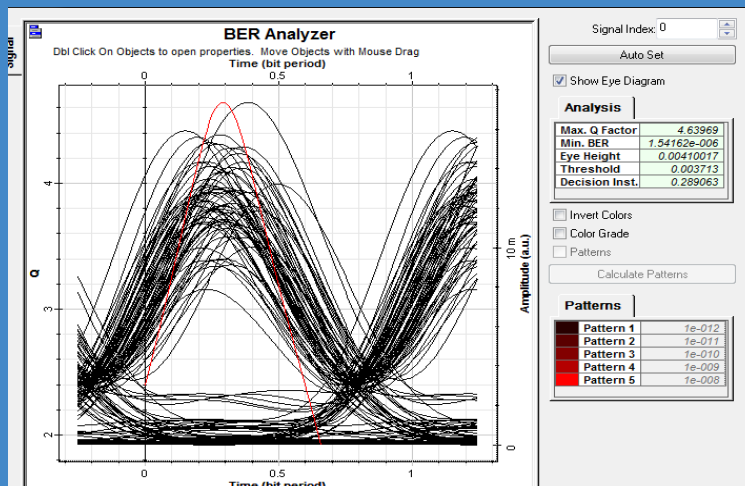
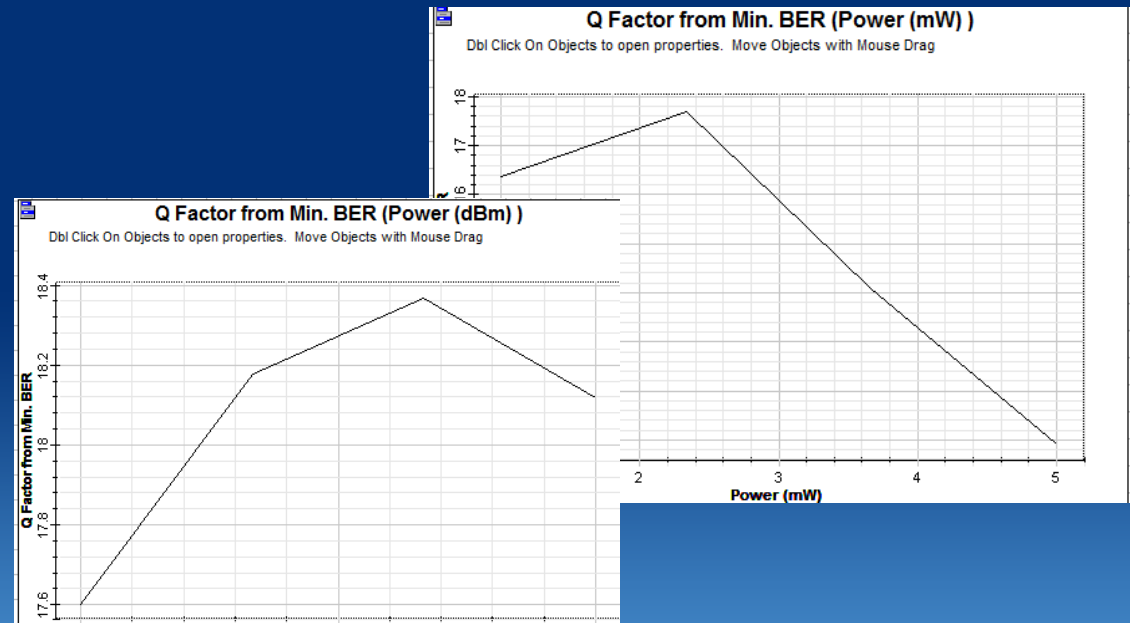
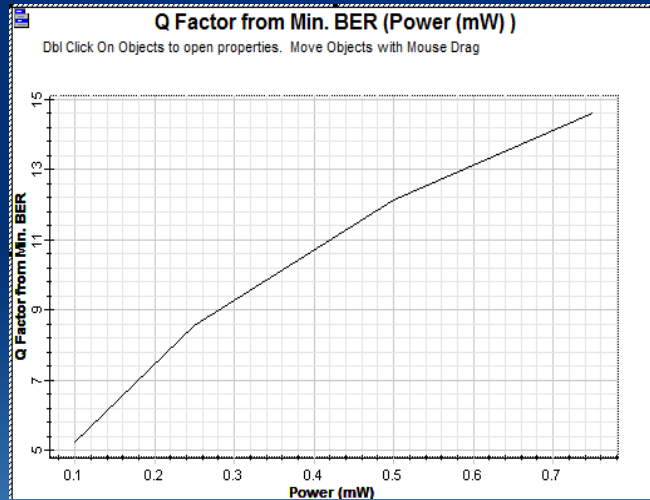


RZ: $Q=18$



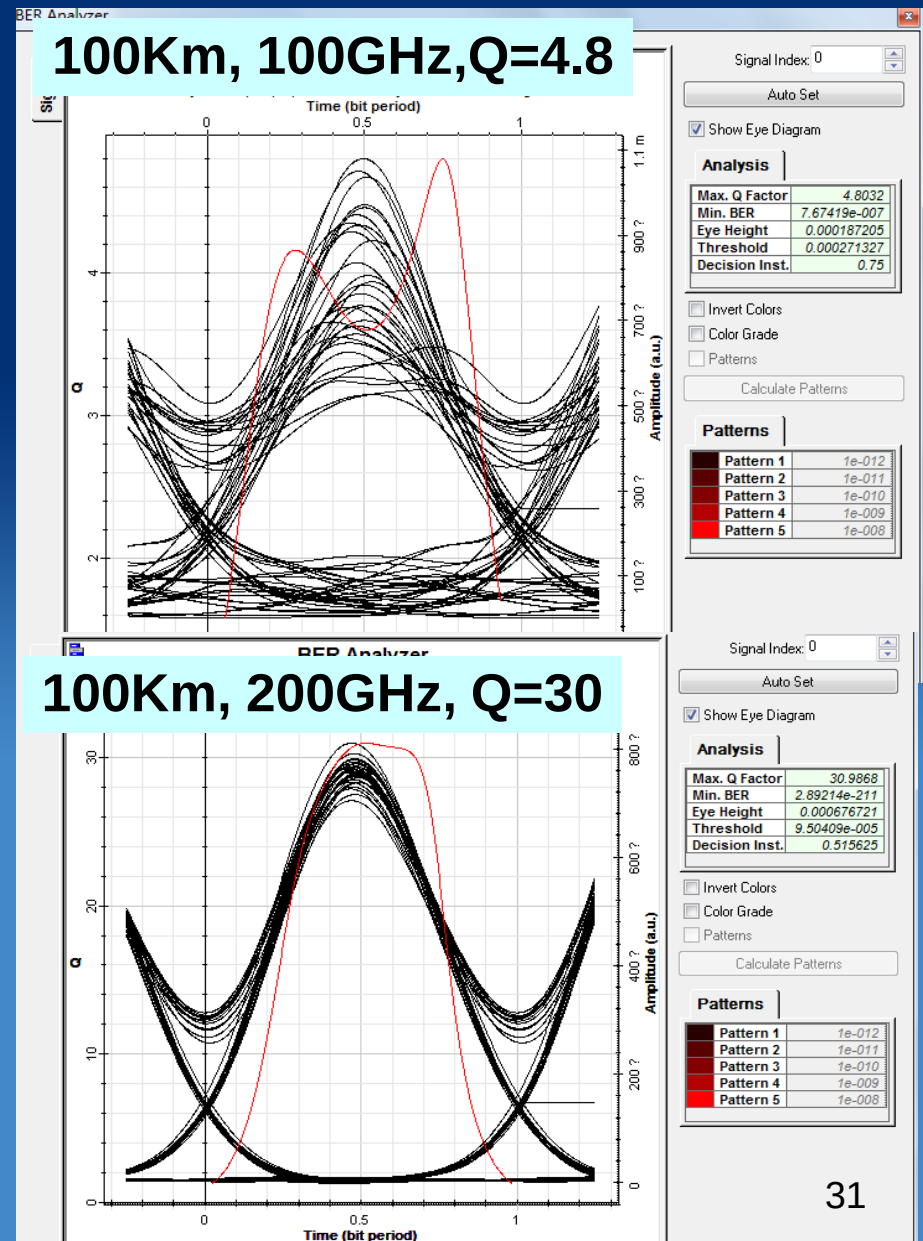
The higher power, the better? Not always!

Q v.s Transmitter power

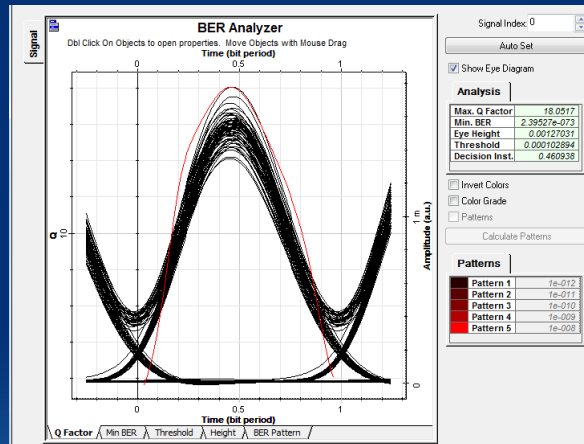


What to do when you find Q is too low?

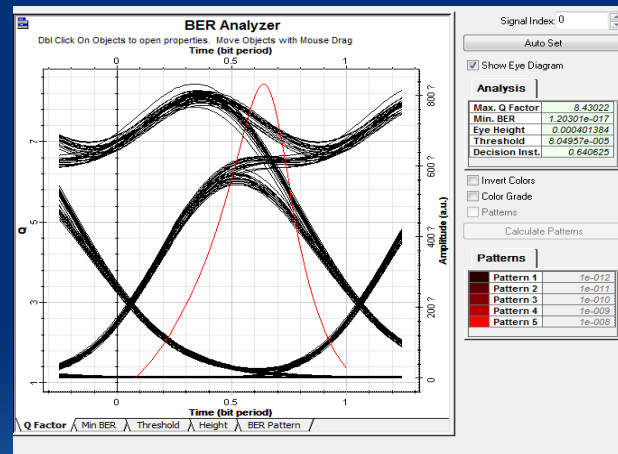
- Cut down distance till it works
- How to increase the distance
 - PMD?
 - spacing 100 or 200GHz



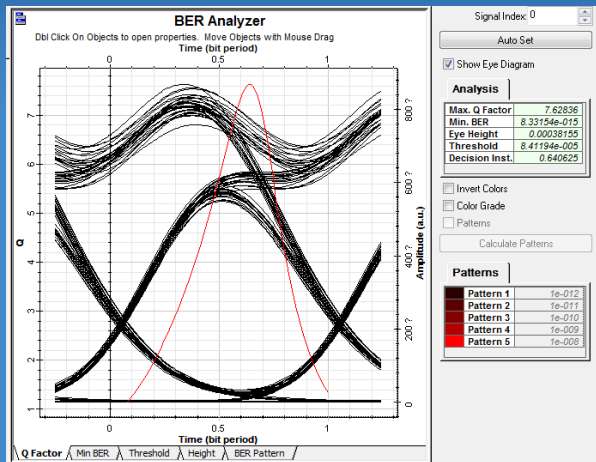
Observe DWDM effect by increasing channel numbers



Single channel, $Q=18$



two channel, $Q=8$



four channel, $Q=7.4$